CRYSTAL SETS TO SIDEBAND

CHAPTER 16C
HOMEBUILDING VHF HAM RADIOS
2 Meter and 6 Meter VHF Receiver Designs

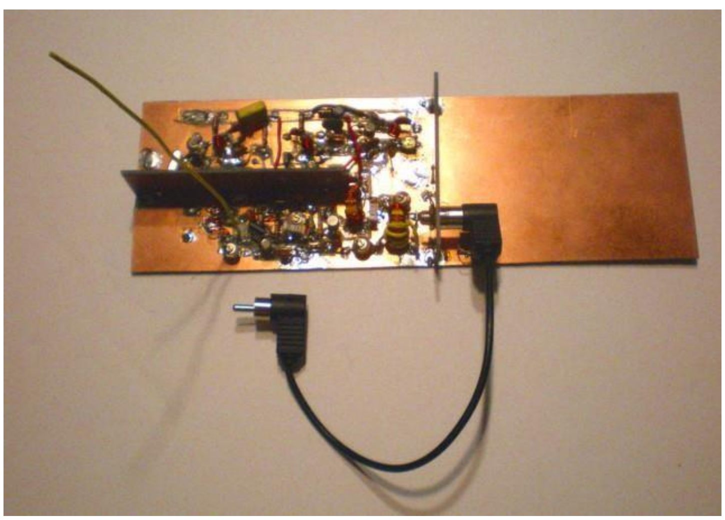
2 meter receiver design approaches

There are several ways to build VHF receivers capable of receiving FM, SSB and AM voice signals. A receiver dedicated just to one band or one frequency could be relatively easy, or at least uncomplicated. Frequency tuning can be as simple as a rotary switch selecting a few crystals for specific frequencies or "channels." As usual, a fully tunable, stable VFO can be the most difficult to homebrew. The obvious choice for me was to build a "converter" to move the VHF signals down to an HF frequency where most of the receiver modules were already developed and working. I used the full range, tunable HF superheterodyne receiver described in Chapter 7C. The 2 and 6 meter bands are 4 MHz wide and the higher UHF bands are even wider. The Chapter 7C receiver is continuously tunable over the entire HF range so it has the capability to cover entire VHF and UHF bands, not just small, 500 KHz portions of them.

If you don't have a full range HF receiver like the one described in Chapter 7C, an easy way to begin might be to use an ordinary FM broadcast radio that tunes from 88 MHz to 108 MHz. Unlike most ham receivers, the broadcast radio already has an FM detector with a VFO frequency locking feature. A converter can move the 2 meter signals down to about 100 MHz where they match the range of the radio. I built such a converter just to prove it works. The audio volume was weak, but that can be improved. The converter circuit was adapted from the VHF to HF converter presented below:

Building a 2 meter to 15 meter converter

For me, two meters is much more useful than 6 meters, so let's start there: My first prototype 2 meter converter is shown below. As you can see, I didn't know how much circuitry would be needed, so I started with a large circuit board.



I found very few articles in ARRL handbooks or on-line to help me. The most helpful receiver article had plans for a converter to move a 2 meter signal down to the 10 meter HF band. It's in the good-ol' 1986 Handbook. One of the stages, the frequency-converting, dual-gate

MOSFET mixer, worked pretty much the way it was described. I modified the output filter to use my usual components and methods, but yes, it worked. It turned out that the rest of the original circuit didn't work at all.

Referring to the photo above, the vertical yellow wire is a temporary antenna for testing. The black coaxial cable with the phono plugs connects the 21 MHz output signal to the HF receiver antenna input jack. 12 volt power enters at the rear of the PC board. The HF receiver has its own 12 volt power supply, so I installed an external jack on the receiver to power the converter.

The entire 2 meter band is 4 MHz wide. The largest HF hamband is 10 meters which is 1.7 MHz wide. Therefore, the usual homebrew approach is convert 2 meters down to 10 meters. Pick the part of the 2 meter band you most want to hear. For example, if you want to receive 145 to 147 MHz, you could tune in most of those frequencies between 28 to 29.7 MHz. To make the conversion you need a crystal controlled reference oscillator. This oscillator needs to be 28 MHz different from 145 MHz. Therefore the crystal must oscillate at 145 - 28 = 117 MHz, or possibly 145 + 28 = 173 MHz. Higher is always harder, so use the lower frequency.

Although I have 10 meter capability in the HF receivers shown in Chapter 13, I don't have a 117 MHz 5th overtone crystal and didn't want to order one and wait 6 weeks. I do have an assortment of VHF crystals that I inherited from a former employer. I rummaged through my collection and decided the best crystal for me was 62.12 MHz. 62 MHz times 2 = 124 MHz. 145 MHz - 124 MHz = 21 MHz. Of course the 4 MHz wide 2 meter band doesn't fit within the 0.45 MHz wide 15 meter band, but that isn't necessary. The entire 2 meter band can received by a full coverage HF receiver tuning from 20 MHz up to 24 MHz.

Eventually, by modifying it stage by stage, I was able to make the entire converter work. The article gave me a way to start the project, so I won't complain. As always, divide and conquer! Don't expect the whole converter to work all at once. Whenever possible, isolate each stage as an independent circuit supplied by simulated signals and monitored using your scope, meters, etc. A circuit lost in the center of a larger circuit is often hard to troubleshoot.

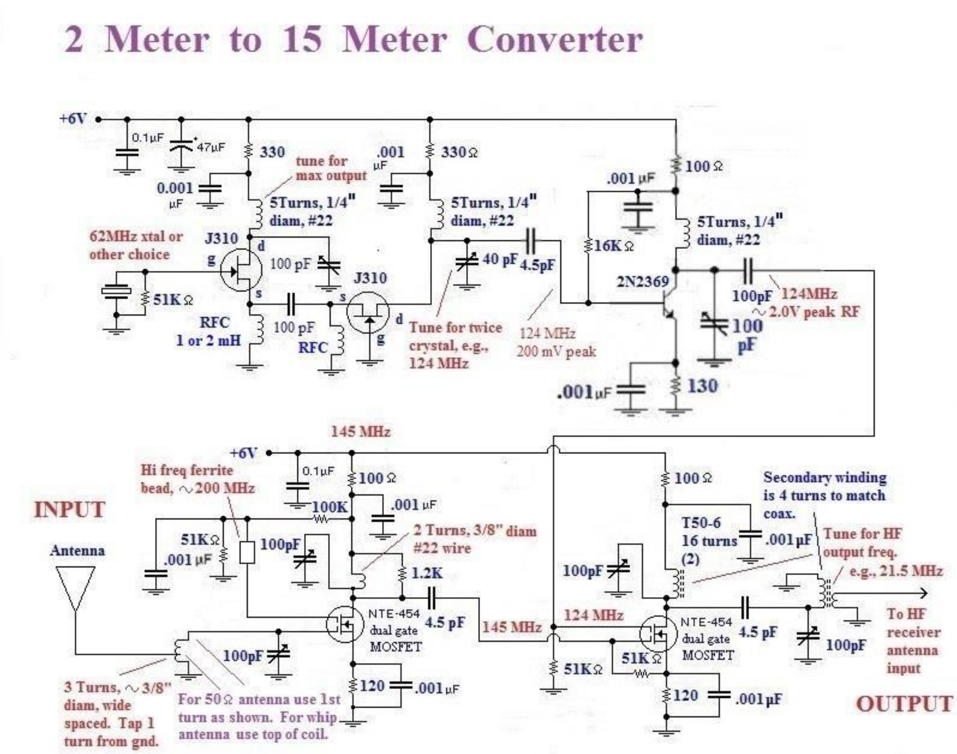
The converter VHF reference oscillator produces the 124 MHz sinewave which is subtracted from the 145 MHz signal. This converter is not really different from the HF converters described in Chapter 13B. You may remember that a dual-gate MOSFET transistor mixer needs a 1 to 2 volt RF signal on the reference gate. Of course the incoming radio signal can be tiny, microvolts peak. To work well, the reference signal must be huge in comparison so that it chops the incoming signal into very definite pieces.

Just like the 2 meter FM generator described in Chapter 16A, I couldn't generate a sinewave larger than the "200 mV peak" my VHF scope indicated. Yes, if you mix a small reference signal with the tiny in-coming signal, you'll produce the desired HF frequency. Unfortunately, the result was faint little carriers barely audible in the HF receiver. Before I was able to amplify the 124 MHz sinewave to more than a volt, the converter output was just a teeny little carrier at 21 MHz barely audible in the HF noise. It was too faint to hear any modulation. In order to finish the converter, I first needed to learn how to amplify VHF RF sinewaves from 200 millivolts peak up to 1 or 2 volts peak - difficult!

I later learned that my measurement of 0.2 volts peak was really more like 2 volts peak. Now I can't explain why the converter didn't work during those first tests.

Complete circuit for the 2 meter converter

Before you build anything, I eventually discovered that all the circuit stages work well at 6 volts and that a +12 volts supply was unnecessary. That ***may*** have been related to some of the circuit failures I experienced. As usual with VHF, single-sided boards are essential.



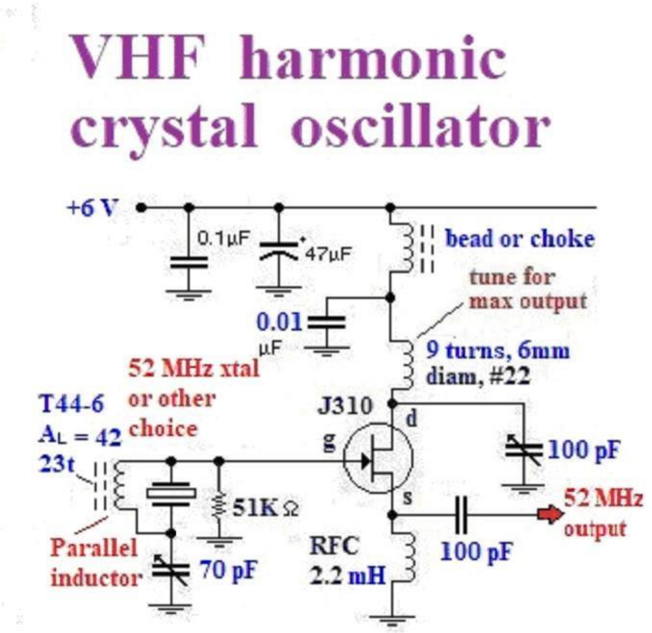
The original ARRL 144 MHz pre-amp connected to the antenna used a JFET transistor with peculiar tuned input and output circuits. I had not seen circuits like that and could not get them to resonate. The dual-gate MOSFET RF pre-amp circuit above was more conventional and worked right away. I believe any similar, small dual gate MOSFET - 3N140, etc - will also work. Also notice that the JFET crystal oscillator circuit uses a signal taken off the source. The original design takes it from the drain. The source signal is *MUCH* larger, but that's because this particular crystal is actually oscillating at 62 MHz, not 20 or 31 MHz. In the ARRL VHF converter designs the output from the second, grounded gate J310 JFET is enough, presumably about one volt peak, to drive the dual gate MOSFET mixer. The laws of physics are different in my basement.

Coaxing VHF harmonic crystals into oscillation

I later learned that it was just dumb luck that the 62 MHz oscillator circuit diagramed above worked so well. The above circuit illustrates the basic approach to harmonic crystal oscillators. An L-C circuit on the drain is tuned to the desired frequency. The crystal is then supposed to oscillate at the harmonic frequency. If you place any harmonic crystal in an oscillator with a Radio Frequency Choke (an untuned) load on the drain, it will oscillate on its prime frequency, 20.7 MHz in this case.

I later tried to build oscillators for 32 MHz and 52 MHz using 3rd harmonic crystals. These crystals were supposed to oscillate at three times their prime frequency. The 32 MHz oscillator insisted on oscillating at 10.67 MHz. The 52 MHz oscillator didn't oscillate at all in the circuit diagramed above. I did some on-line research and noticed that VHF harmonic oscillator circuits almost always have an inductor in parallel with the crystal. My impression is that ***3rd and 5th harmonic crystals will rarely oscillate above 50 MHz without a parallel inductor.***

Below is the diagram of a working 52 MHz oscillator:



An inductor in parallel with the crystal cancels the crystal capacitance.

A quartz crystal is built like a simple capacitor. It consists of two parallel metal plates with a quartz "insulator" in between the plates. One can imagine that the invention of quartz crystals was an accident by some fellow trying to find new high voltage insulators for his capacitors. I always assumed that the simple capacitance was part of the resonance L-C. I suppose that's true until the metal plate capacitance becomes larger than the quartz L-C equivalent capacitance.

I measured the capacitance of my size **HC-49 VHF harmonic crystals**. They all had capacitances in the range of **3 to 4.5 pF**. The main function of the inductor is that ***it should resonate with the capacitance*** of the crystal and make this larger capacitance vanish. In other words, the crystal assembly becomes two L-C s in parallel. No wonder it's tricky to make this work!

Using our usual inductor calculations, wind an inductor that will resonate with the 3 or 4 pF capacitance at the crystal frequency. If you do that, I can almost guarantee that it will oscillate, although it may not be stable and it may not lock into one frequency. I had to fiddle with the number of turns on the parallel inductor to get it to lock on a frequency near 52 MHz.

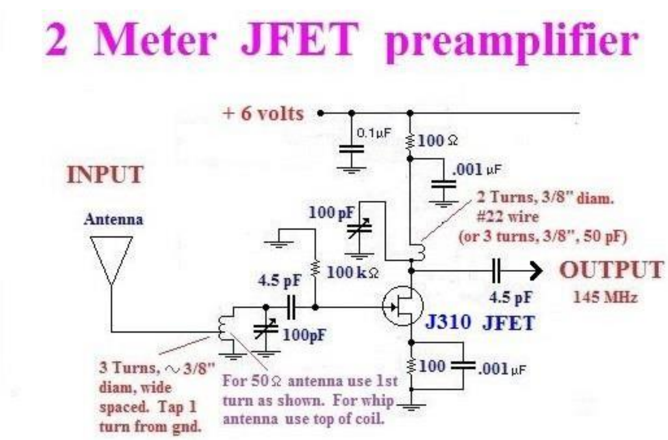
Finding the optimum parallel inductor

So far, I have built 3 of these VHF 52 MHz crystal oscillators with parallel inductors. Each one was different! The inductors needed 32, 23 and 12 turns using T50-6 or T44-6 toroid cores. (The A_L multipliers of these two toroid cores are nearly the same, 40 and 42, respectively.) I finally concluded that I should begin by finding the parallel inductor that makes the crystal L-C circuit oscillate at about the right frequency. Next, tune the other parallel L-C in the drain circuit to adjust the frequency to the precise crystal frequency. It should lock-in solidly. Confirm this by raising and lowering the supply voltage, say between 5 and 12 volts. The frequency shouldn't change more than a KHz or so.

Finally, look at the form of the sinewave on your oscilloscope. The sinewave should be uniform and symmetrical. If the negative half-wave looks short and fat while the positive half sinewave is more spike-like, the adjustment is not optimum. Tweak the two trimmer capacitors back and forth, until it suddenly looks symmetrical. The exact frequency will probably drop down a hundred KHz as well.

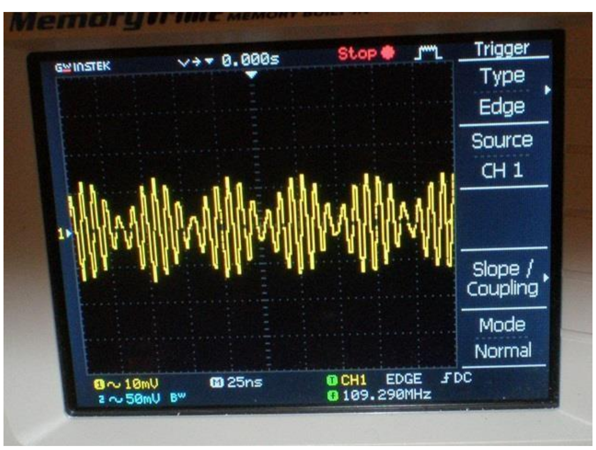
A JFET 2 meter preamplifier

The complete converter circuit presented above uses a dual-gate MOSFET preamplifier. If you prefer, here's a JFET circuit that also works well:



Adjusting the converter HF output L-C s

Tuning the L-C s in the converter can be difficult. As always, divide and conquer. Start with the 124 MHz reference waveform oscillator stage then work up the stages to the input to the mixer dual MOSFET. If possible, figure out a way to inject a 21.5 MHz signal generator signal into the last two tuned L-C s. These two L-C s are quite precise - hi Q. If one of them is slightly out of tune, there will be little or no signal at the output. Finally, expose the converter to a strong 2 meter signal. Tune the two 144 MHz resonant circuits in the RF input amplifier while observing the 21.5 MHz output. When the 2 meter input is properly tuned, a strong 21.5 MHz sinewave-modulated 124 MHz sinewave will appear at the output.



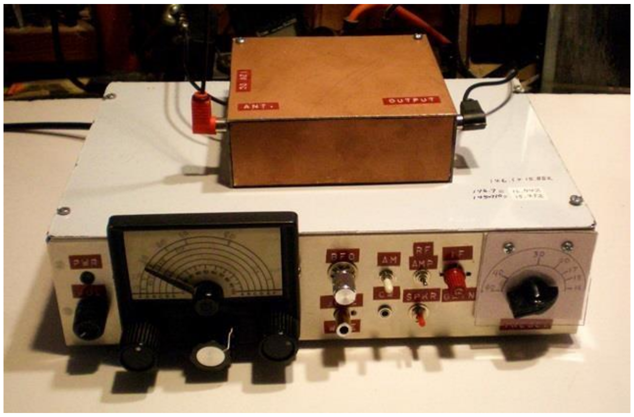
Here's the converter output while receiving a 2 meter signal. The burst modulation is the final filtered 21 MHz component that comes out of the mixer. 145 MHz - 124 MHz = 21 MHz. There are 6 sinewaves per burst. Notice that 124 MHz ÷ 6 = 20.7 MHz. The 109 MHz read-out is incorrect because the counter can't detect the smallest sinewave cycles. When there is no 2 meter signal, the output waveform is just the pure 124 MHz sinewave reference.

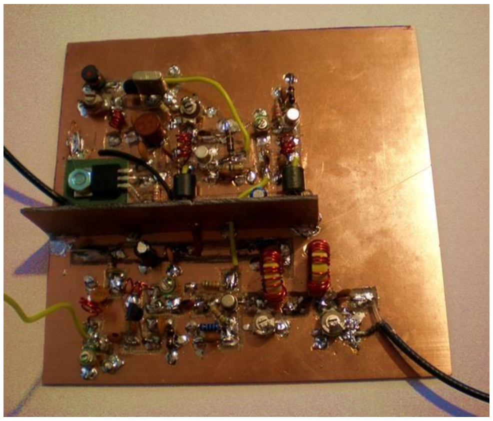
I was surprised to discover that the converter worked well with as little as a 5 volts DC supply. Yes, the mixer needs a "big" reference frequency input sinewave, but it turned out that "big" is proportional to the supply voltage. So if the supply is half as high, one half volt of reference signal is enough. 6 volts has the advantage that the converter draws 1/4 as much DC power. I eventually added an LM7806 linear DC regulator chip to the converter board to gracefully reduce the 12 volt supply to 6 volts.

Tuning 2 meters with an HF receiver

Although it could receive real 2 meter signals with good sensitivity and audio quality, tuning in the signals using the Chapter 7C superheterodyne required patience. The preselector, the RF gain, IF amplifier gain, the audio gain and the band-width switch and all had to be selected, adjusted and balanced to get ideal reception. When out of tune, it produced squeals, howls, loud static noise, silence or speech distortion. Because I'm accustomed to working with un-enclosed bare boards, it took me ages to realize that the Chapter 7C receiver only works smoothly for this application when fully enclosed in its aluminum cabinet, especially the top cover. In contrast, the converter seems to work the same as a bare board or packaged in the shielded circuit board box shown below.

When the converter was operated as a bare board, I used a piece of wood to separate the single-sided boards from the solid, grounded metal box. This kept the metal receiver lid from detuning the VHF circuits. Later, when enclosed in a box made from circuit board material, I used a 1/2 inch of plastic foam inside the copper-shielded box for the same purpose.





In the photo on the left above, the 2 meter converter is shown with a 46 cm high piece of stiff black wire for an antenna. As explained earlier, I can hear my local repeater and VHF signal generator using this crude set-up. However, the signal amplitude and audio quality varied wildly when I place a hand on the receiver enclosure or brought my hand near the antenna. Eventually I replaced the short wire antenna with coax connected to my roof-top 2 meter vertical antenna - Big improvement! The roof-top antenna is a rugged, outdoor version of the same 46 cm vertical ground plane antenna. It uses the same formula-calculated design as those described in Chapter 5B for the large 20 meter through 10 meter verticals. The circuit board is the 3rd version of the converter - hopefully the final version.

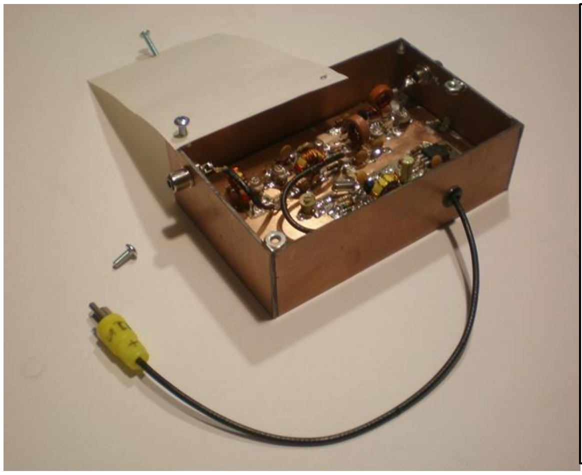
Even with the un-enclosed 2 meter converter, the receiver now acts smoothly without loud noises and precise adjustments. Now I can put my hands all around the converter with no change in reception.

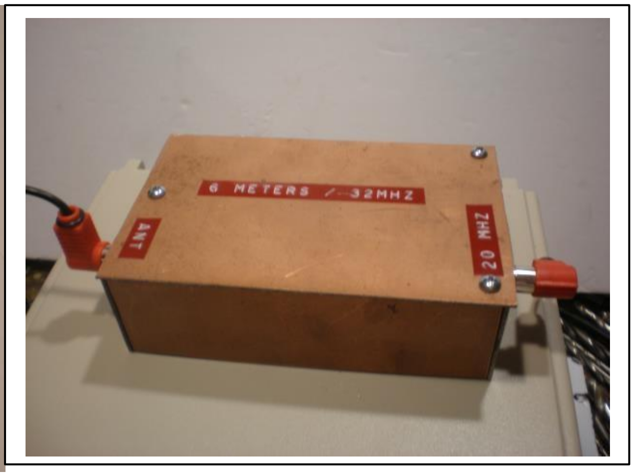
The copper box enclosure

The copper box is made from 2-sided PC board. The purposes of the box are shielding from ordinary 21 MHz signals and appearance. The box is soldered together inside at a few strategic points. I didn't attempt to solder all the seams. I had previously learned that PC board warps when the seams are completely sealed with solder. The top is held on by three 6-32 machine screws. With care, 6-32 nuts can be securely soldered to the corners of the box. The circuit board was placed on a foam pad inside the box. The foam keeps the VHF circuitry away from ground so the VHF signals will not be overwhelmed by capacitance to the grounded metal case. I connected the PC board ground traces to the inner wall using a wide piece of solder wick - a woven copper strap. There are holes in the walls of the box for the RF and power connectors to pass out of the box. The last wall of the box had to be soldered in place after the circuit board was installed.

In order to solder the steel 6-32 nuts, it's necessary to tin them first. Tin-lead solder sticks poorly to steel so I applied a tiny speck of plumber's acid flux to the edge of the nuts. The solder should cover about half of the edges of the nuts. The locations for the nuts on the box were also tinned. While holding the nuts with a hemostat, I carefully heated the tinned solder locations to fuse the nuts in place.

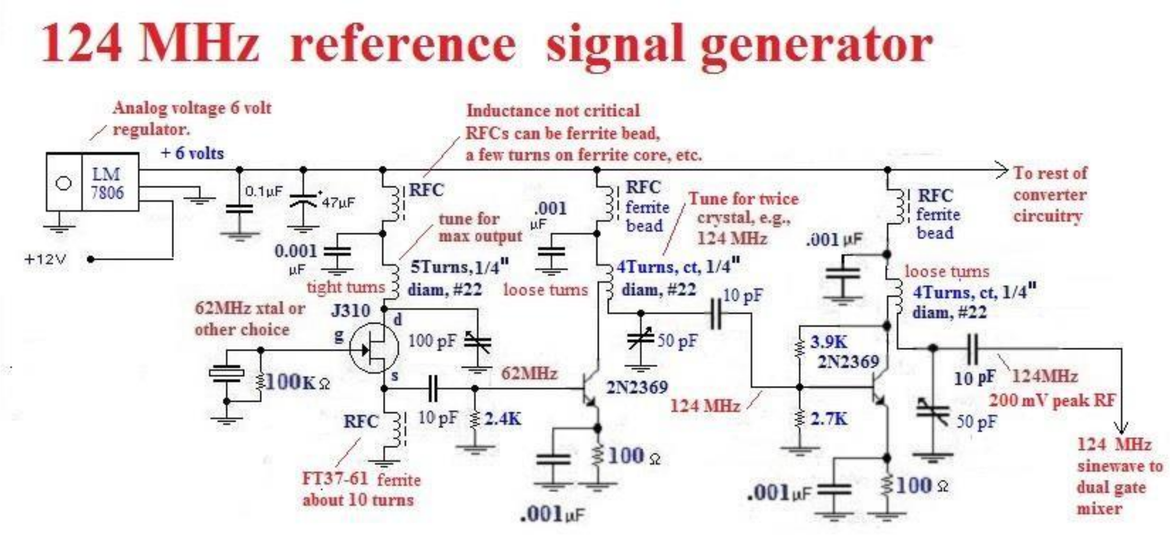
This is probably obvious, but I needed to figure out the precise location for the holes on the lid to align with the 6-32 nuts. I made a paper pattern replica of the lid, then placed it on the box. Using a sharp pencil point, I poked holes in the paper aligned with the 6-32 nuts. Next, I laid the paper pattern on top of the P.C. board lid, marked the locations for the holes on the lid and drilled them. The pictures below illustrate the process on a 6 meter converter.





Like all my VHF projects, the 2 meter converter soon stopped working and had to be repaired. As planned, I was able to disassemble the box and replace the circuit board without much difficulty.

Many months later: Long after the original converter circuit began working, the 124 MHz reference generator died. I replaced the J310 JFET doubler and 2N2369 transistors, but the reference generator circuit remained too feeble to function. The VHF gremlin had struck again! After several repair attempts, I gave up and added a new, large P.C. board that I simply bolted onto the old board. The new working circuit was unnecessarily complicated, but I was able to trim it down to the circuit shown below:



This circuit produces a robust, reliable reference signal - at least it has for many months. The inductance of the decoupling inductors or ferrite beads is not critical. The inductance just needs to be several microhenries, a value unlikely to resonate at the VHF frequency. Similarly, the RFC on the oscillator source needs an inductance at least 10 or 20 times the inductance in the tank L-C circuit. The resistance of the wires in the inductors should be essentially zero so that the output voltage swing is not reduced.

I used the tapped tank circuit design for the amplifiers. You may remember from Chapter 11 that tapped inductor outputs are excellent for filtering out unwanted frequencies and produce big output voltages. The disadvantage is that the output is very high impedance. There can be hardly any output current. Fortunately, the input of a dual gate mixer is extremely high impedance and doesn't need significant drive current.

Wait! Why does the scope only read 200 mVolts peak instead of one or two volts peak? As was explained in Chapter 16A, I eventually learned that my "VHF scope" reads only about 1/10 of the real VHF voltage! The real voltage of the original generator must have been more like 10 or 12 volts peak? The dual gate mixer only needs roughly one volt to saturate with each sinewave pulse. Apparently ten times more than that didn't hurt the mixer operation. Now the mystery becomes, why didn't "200 millivolts peak" work when I first built the converter?

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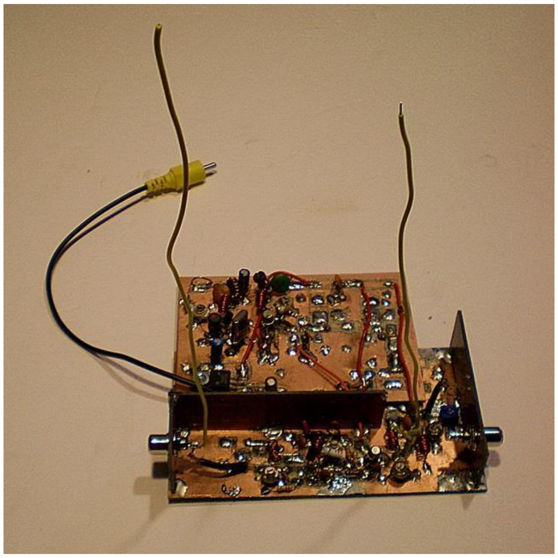
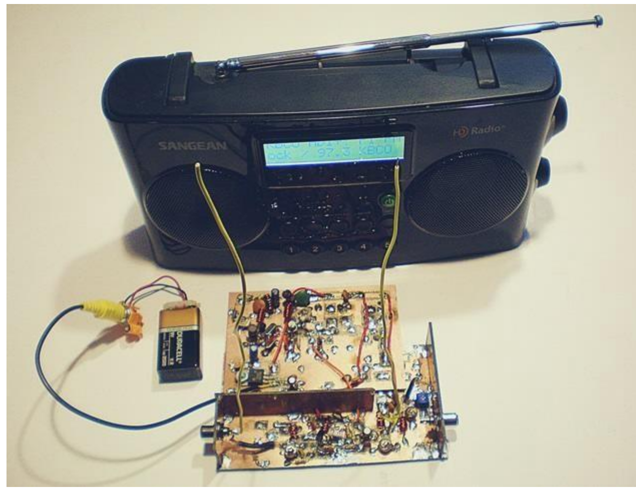
Listening to 2 meters with an FM broadcast radio

After I had refined the design for the 2 meter to 15 meter converter described above, I wondered if I really could convert a broadcast FM radio to 2 meters. The sound volume is too weak, but yes! It works.

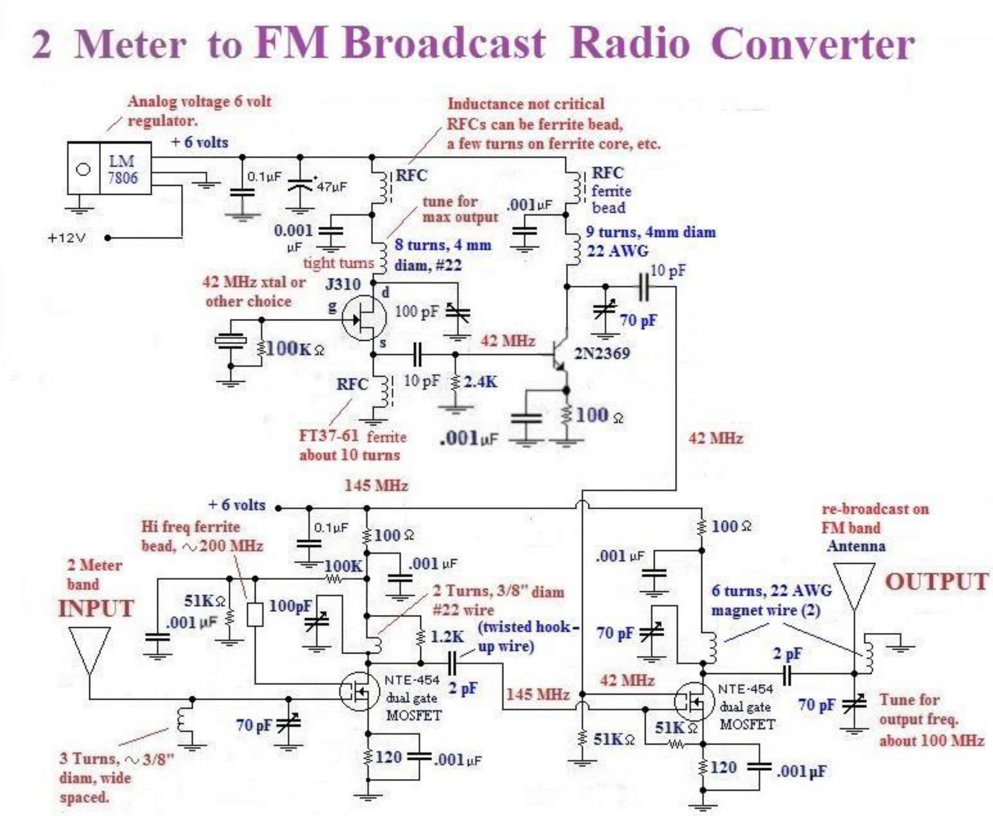




Here is the complete experimental converter: The boards and components were salvaged from my first 2 meter converter board(s). The vertical yellow wire on the left is a crude 2 meter antenna. The yellow vertical antenna on the right re-broadcasts the converted 2 meter signal within the FM commercial band, 88 to 108 MHz. To use it with an unmodified FM radio, the converter is simply placed next to the FM receiver. I'm not kidding! A direct connection with the FM radio antenna would be ideal but is NOT necessarily needed.

The prototype is set up just as shown on the right. A 9 volt battery is enough to power it. The audio is weak, but is perfectly clear. This happens because the commercial receiver is designed for super-wideband FM, +/- 75 KHz! In contrast, ham radio FM is supposed to be limited to +/- 5 KHz. To generate an FM modulated 2 meter test signal, I used the 2 meter signal generator with a walkman radio for an audio source. This was described at the beginning of Chapter 16A.

When there is no background noise, the receiver volume can be comfortably turned up to maximum. As always, the converter L-Cs in each stage must be carefully aligned for good sensitivity. Like my other converters, I can receive a strong signal from 3 meters away from the 2 meter signal generation board. The final amplifier was not needed for the test.



I used a 48 MHz reference crystal. It worked without a parallel inductor and tuned up using the same 62 MHz oscillator circuit shown earlier. Only one stage of amplification was needed. The L-C circuit for the 48 MHz amplifier was a 70 pF trimmer in parallel with an 8 turn, 4 mm diameter, 22 AWG, close spaced coil. Unfortunately, the 2nd harmonic of 48 MHz is 96 MHz and was overwhelmingly strong around that frequency. The 2 meter signal generator was set to 145.71 MHz which was converted to 97.71 MHz. This was more than far enough from 96 MHz to avoid interference.

The ideal oscillator crystal would be 40 to 43 MHz. 148 MHz - 40 MHz = 108 MHz and 144 MHz - 40 MHz = 104 MHz. Two times 40 MHz = 80 MHz, which would not interfere with the frequency range of the FM receiver. The only other change I needed was to replace the 21 MHz inductors in the output with 100 MHz coils. They were 6 turn, 4 mm diameter, close-spaced coils made from 22 AWG magnet wire. The coupling capacitor between the two L-C circuits is simply two pieces of insulated hook up wire twisted together, one turn. The coils were tuned to the desired FM receiver frequency.

A serious FM radio conversion

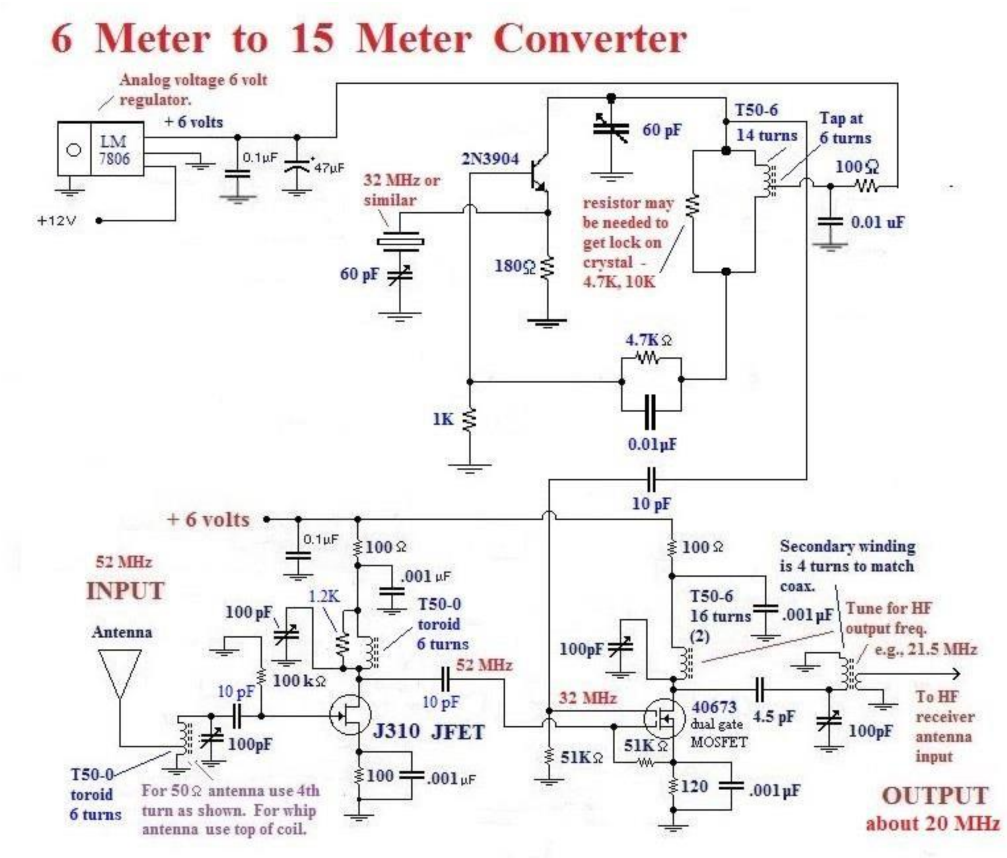
To make a practical 2 meter receiver out of this, the FM radio would need to be taken apart and placed inside a metal enclosure with the converter. This would allow full use of the FM receiver range and filter out the commercial FM stations. It would also be easy to add an additional stage of audio amplification. This would be a mechanical challenge because we would still need access to the controls and frequency display while shielding out the FM broadcast stations.

Thin, clear plastic sheets are available that have an invisible layer of conductive tin oxide on them. This material can be placed over a window in a metal enclosure to protect the frequency display from RF interference. Once at work I had to protect the digital displays of medical diathermy generators. With no conductive shielding, the high power 434 MHz and 2.45 GHz radio waves made the digital circuitry go nuts. Before I found the clear conductive plastic shielding, the problem had me totally stymied.

A 6 Meter to 15 Meter Converter

My interest in 6 meters was inspired when one of the guys on our 2 meter round table told us about working stations on 6 meters during a brief opening that same day. He told us that, although FT-8 was most of the ham activity, there were also stations on upper sideband.

Below is a 6 meter converter based on the same circuits I developed for 2 meters. My goal was the same as the 2 meter receiver: The 6 meter band is 4 MHz wide and I wanted to tune all of it. Years ago I built a 6 meter converter for the receiver described in Chapter 13. It only tunes the bottom 500 KHz. I now realize that the lower 500 KHz covers most of the 6 meter activity, so it's usually adequate. The Chapter 13 receiver has variable crystal filter bandwidths and an RF input attenuator. These features make it easier to tune in single sideband than the simpler Chapter 7C receiver.



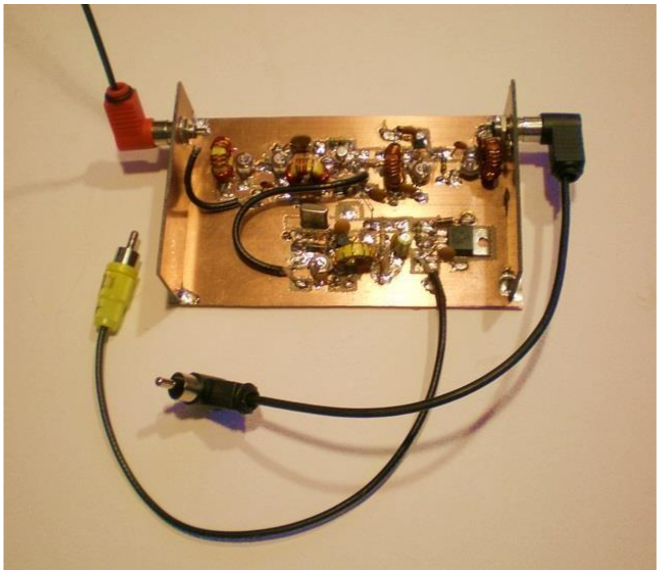
This new converter was designed to work with the Chapter 7C receiver. The preamplifier and mixer circuits are the same as the 2 meter converters shown earlier, but uses a 32 MHz reference oscillator to move 6 meters down to approximately 15 meters. The reference frequency - about 30 MHz - is subtracted from 50 MHz to produce a 20 MHz signal. The Chapter 7C receiver can tune continuously from 80 meters to 30 MHz, so it can receive the entire 6 meter band moved down to 20 to 24 MHz.

The main difficulty I had getting the converter to work was that the 32 MHz 3rd overtone microcomputer crystal refused to oscillate at 32 MHz. I used the same oscillator circuit I had used for the 2 meter converters with L-C circuits redesigned for 32 MHz. As explained earlier, no matter what I did, the crystal insisted on oscillating at its primary frequency, 10.068 MHz. I gave up and rebuilt it using the same Butler oscillator used in most converters and transmitters described in earlier chapters. I happened to have a 28.2 MHz primary frequency crystal and it worked immediately. The oscillator output alone was sufficiently high and no amplifier was needed.

Later I experimented and put in a parallel inductor across the 32 MHz crystal. I used the same oscillator design described for the 52 MHz crystal oscillator. Although it required some fussing with the number of turns on the parallel inductor, it eventually locked solidly at 32 MHz.

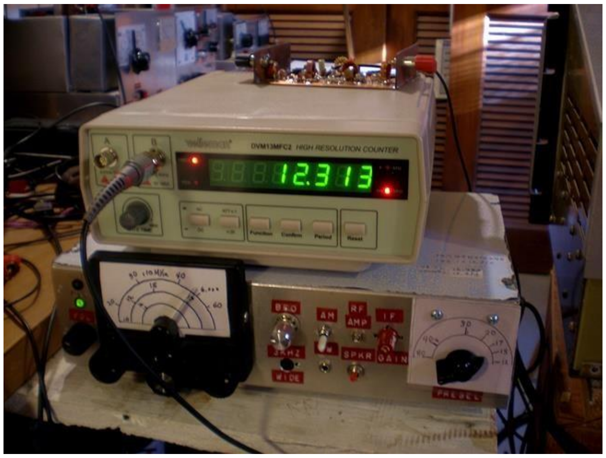
I raised the 6 meter preamplifier input and output capacitors from 4.5 pF to 10 pF. This raised the gain, but was plagued with oscillation. I cured this a 1.2 K resistor across the amplifier drain coil. Tuning up this preamp still requires trial and error. The optimum settings are a trade-off between oscillation and sensitivity. When properly adjusted, it works very well.

The 40763 dual gate MOSFET was purchased from RF Parts Company and seems to work as well as any of the other dual gate MOSFETs. At $7 each they are still expensive, but they have wire leads and are much easier to mount than the tiny surface mount MOSFETs. The main virtue of MOSFET mixers is that, without any PN junctions, they are very low noise compared with ordinary bipolar transistors.



The 6 Meter to 15 meter circuit board

The converter prototype shown above can receive a 50.15 MHz SSB signal from a breadboard SSB driver across the room as described in Chapter 16D. Speech is totally intelligible, but there is some distortion on loud voice peaks. I believe this small defect is in the homebrew transmitter, not the receiver. The degree of sensitivity resembles other converters described earlier. The red connector goes to the 6 meter antenna. The yellow connector is the +12 volt power connection and plugs into the receiver power supply. The black connector goes to the receiver antenna input. The 6 meter conversion in operation is shown below. The frequency counter makes tuning the VFO much easier.



Another difficulty is that, as usual, the 6 meter band is stone dead. Because 6 meter band openings are rare these days, another project would be a 6 meter band signal detector that would tell us when we should rush over to the ham shack and listen. One suggested frequency for FT-8 is 50.313 MHz. Perhaps I should just tune the receiver there and leave it with the audio turned up. The loud whining noise would tell me when the band was open. I can precisely tune the Chapter 7C receiver using a frequency counter to monitor the VFO. As you can see, arithmetic is needed to derive the correct HF listening frequency and the VFO frequency to reach that frequency. I installed a connector on the back of the Chapter 7C chassis to make monitoring the VFO convenient.

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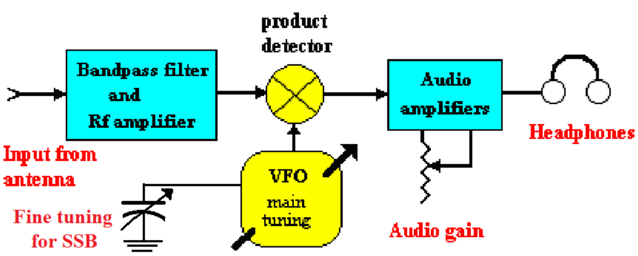
A DIRECT CONVERSION SSB / CW 6 METER RECEIVER

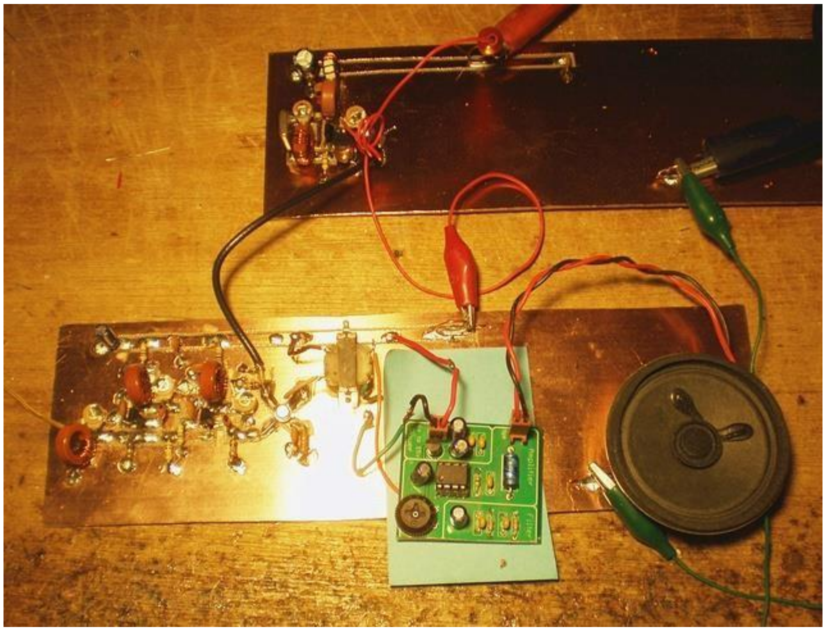
If you don't have an full range HF receiver like the one in Chapter 7C, a direct-coupled (DC), dedicated 6 meter SSB receiver may be a practical homebrew approach. In general, "DC" receivers can be much simpler than building a superheterodyne. DC receivers receive SSB or CW without IF amplifiers and additional detectors. They can also receive narrow-band FM. (See Chapters 7A and 16A.) Unfortunately, DC receivers are not simple for 6 meter SSB because, ideally, they need a **stable**, finely tuned VFO that tunes 50 to 54 MHz. Unlike wideband FM, tuning in SSB requires precise frequency tuning within a hundred Hz. In contrast, wideband 2 meter FM can still be understood 3 or 4 KHz off frequency.



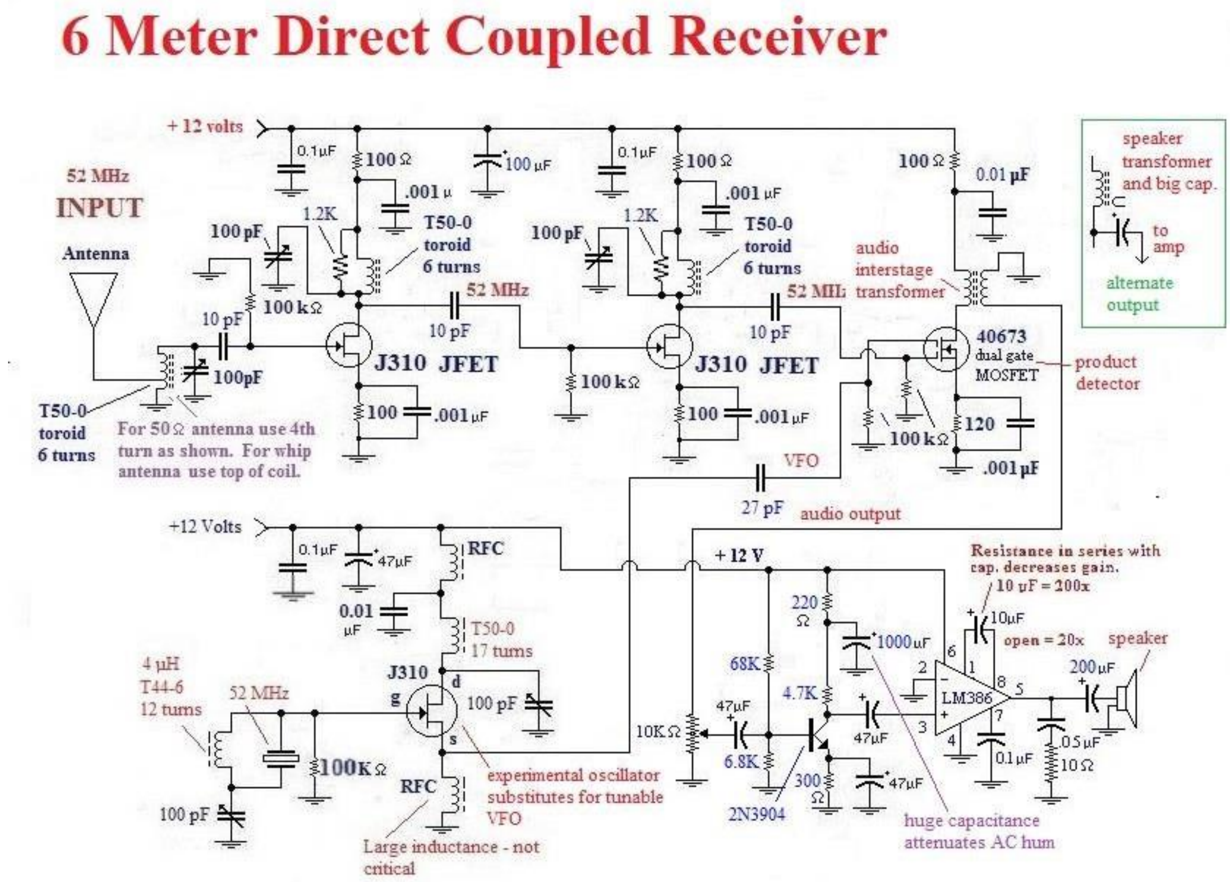
6 Meter Direct Conversion Receiver

This receiver project investigated whether a DC receiver is practical at 51 MHz - **it is!** Rather than go to the effort to build a tunable, stable 6 meter VFO, I used a crystal controlled 51 MHz oscillator for the local oscillator. The photo below shows the working 6 meter DC receiver. It can receive the AM signal from the QRP AM transmitter described in Chapter 16D. Like all, DC receivers, they are good for SSB or CW, but must be tuned perfectly to receive AM without an annoying whistle on the audio. When I set up this experiment, I wasn't convinced I could build such a thing. I had doubts because both the transmitter and the receiver must be extremely stable to generate and tune in SSB.

The VFO in the 6 meter SSB transmitter described Chapter 16D has a stable VFO. But considering the rare 6 meter band openings, I probably wouldn't find any SSB speech stations to listen to. For the experiment I used a crystal controlled oscillator as a temporary VFO. By luck I was able to make the second 51 MHz crystal oscillate on the same frequency as the little 6 meter AM transmitter described in Chapter 16D. Using the crystal oscillator "VFO," the DC receiver was able to receive the AM transmitter with good fidelity and gain. The crystal oscillators proved to be stable enough to tune out the annoying audio side-tone whistle.



The upper board just has the 51.3 MHz local crystal oscillator. I built it separately because I eventually hoped to use the VFO described in the next article as the tuning device. The lower board has the antenna (the yellow vertical wire), two stages of 51 MHz preamplifier and the product detector. The green commercial circuit board is an audio amplifier from an FM receiver that I salvaged. It has the same old LM386 amplifier circuit used several times in this book. The light green Post-It-Note paper insulates the LM386 board from the bare copper surface. The circuit diagram is shown below:

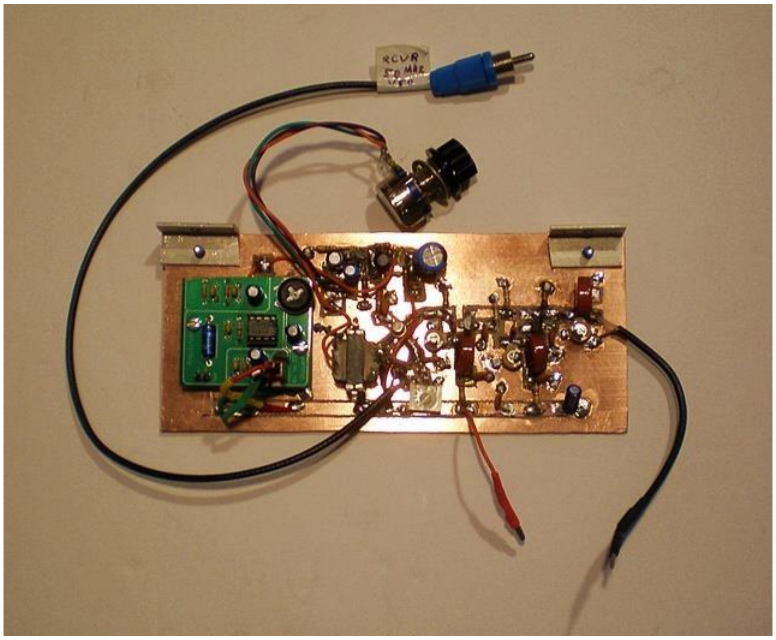


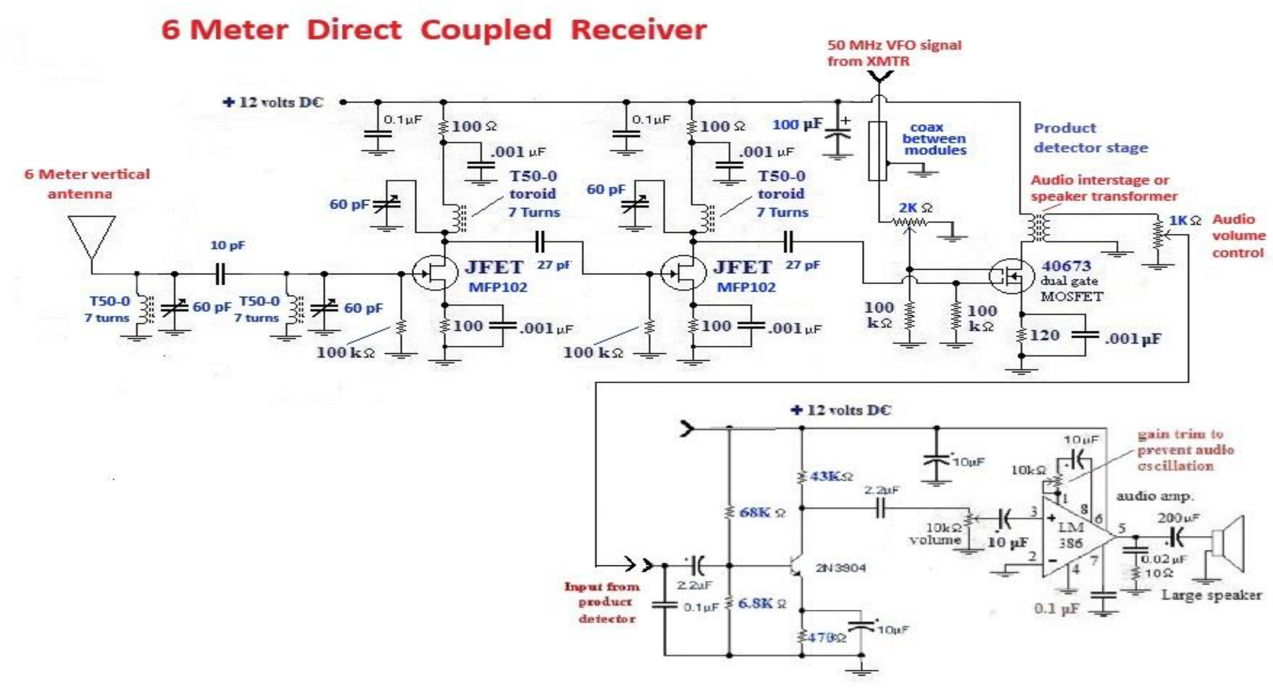
I added a second 6 meter RF preamplifier because there is no IF amplifier strip to increase overall gain. As shown in the photo above, I was surprised to discover that an extra stage of audio amplification wasn't needed. Later, when I installed this receiver in the 6 meter transceiver described in Chapter 16D, the speaker was pointed to one side and was muffled by the enclosure. In that application the extra, single-stage, transistor audio preamplifier restored the volume. When you first build the extra amplifier, check to see that the collector resting voltage is between 5 and 7 volts. If not, raise or lower the 68K bias resistance to correct this.

The difference frequencies between the input 6 meter signal and the 6 meter VFO signal are the low, audio frequencies. Because they are low frequency, the product detector transformer must be high inductance. I used a small audio interstage transformer. Both windings are high inductance. You could also use the high inductance winding of a small speaker transformer and a large electrolytic capacitor as shown in the alternate circuit.

Final Design for the DC receiver

The photo below is the 6 meter DC receiver board ready to be installed in the 6 meter transceiver. By the time I had the circuit tweaked and adapted for use with the VFO, it had changed somewhat. The point of showing the final circuit is to emphasize that there are dozens of circuits that can do this job. Lots of component choices and values will work. Just keep fixing each stage of your circuit until you're happy with its operation.





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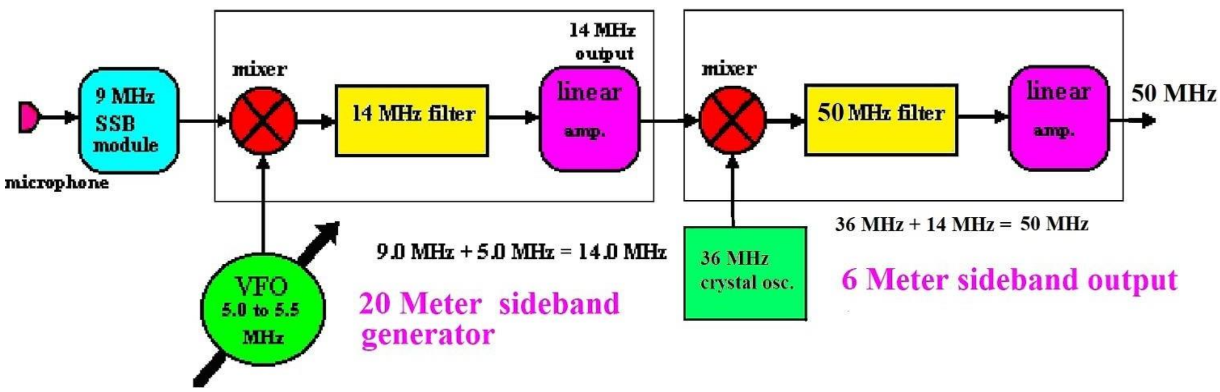
A tunable 6 meter VFO

If you jump ahead to Chapter 16D, you'll see that the 6 meter SSB transmitter demonstrates how to build the necessary 6 Meter sinewave VFO. The Chapter 16D transmitter begins with a 5 MHz VFO (Chapter 10) then adds it to the 9 MHz SSB signal. This generates a 14 MHz SSB signal. The 14 MHz signal is added to a 36 MHz sinewave to produce the 50 MHz SSB signal. A block diagram of the Chapter 16D transmitter SSB generator is shown below. Replacing the 9 MHz SSB generator with a 9 MHz crystal oscillator converts the 6 meter SSB generator into a 50 MHz VFO that works well with the 6 meter DC receiver described above.



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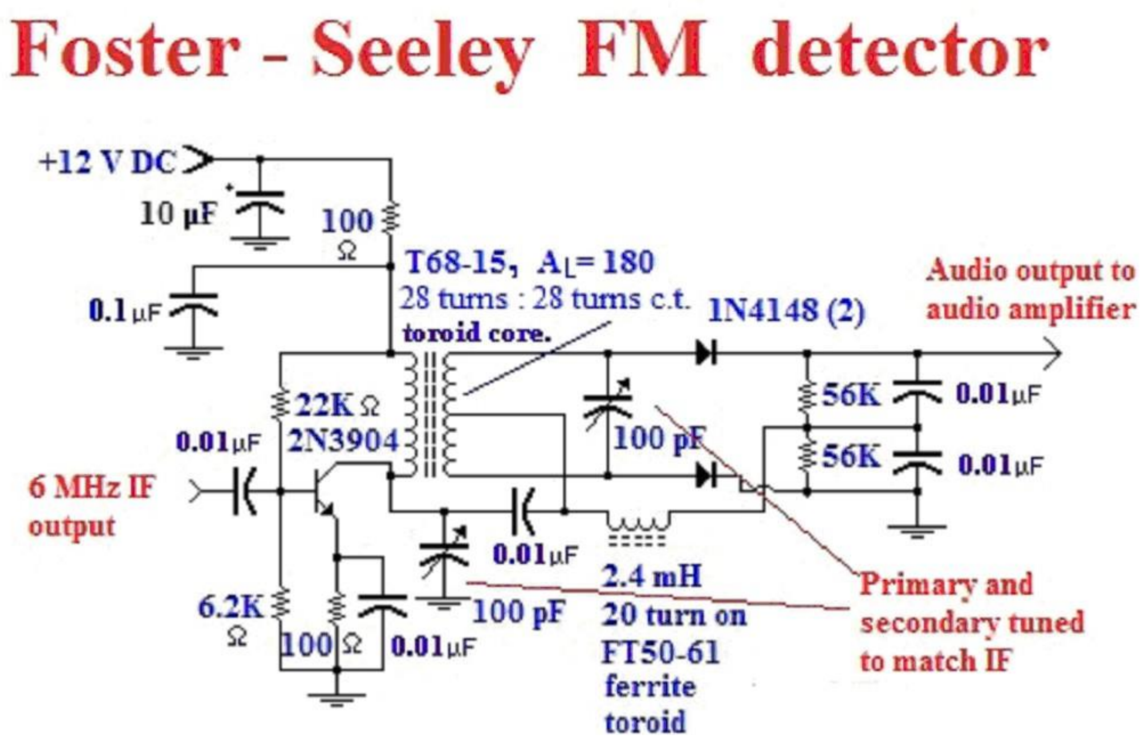
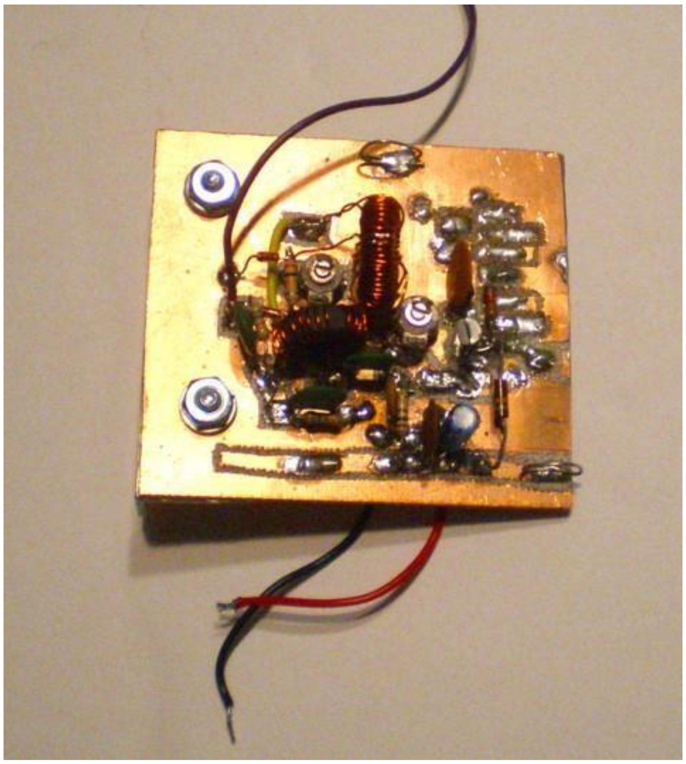
A Foster-Seeley FM Detector

A difficulty with using the HF receiver described in Chapter 7C was that it didn't have an FM detector for FM signals on 2 meters or elsewhere. I could hear a strong carrier on 21.45 MHz, exactly where the converter was supposed to move the 2 meter signal. However, with only an AM detector or the product detector, I heard no audio. So my next project was building a real, wideband FM detector for the Chapter 7C receiver. The receiver described in Chapter 13B has a discriminator FM detector, but I didn't want to be restricted in tuning range on 2 meters. The fun of this VHF project is experimenting with circuits new to me, so I looked for a description of an FM ratio detector. On-line I found the Foster-Seeley FM detector circuit which is similar to a ratio detector but looked simpler. I had never heard of a "Foster-Seeley," so I built one.



It consists of an additional single IF amplifier stage feeding a resonant transformer primary. The secondary winding is also tuned to the IF frequency and is rectified as if it were a full-wave rectifier output. The output is a DC signal. The DC varies in amplitude audio cycle by cycle. In other words, it converts varying frequency to varying DC amplitude. Unlike a product detector, this is a true FM to audio converter. The magic happens when the transformer secondary output is combined with an out-of-phase IF signal derived from the primary side of the transformer. The two signals cancel and reinforce each other, depending on whether the frequency rises above or falls below the exact IF frequency. The resulting out-of-phase signals are rectified by the full wave rectifier. Because the out-of-phase signal is introduced twice, before and after the inductor, exactly how the interference produces a uni-directional DC signal is abstract and I don't fully grasp it. My best guess is that signals above the IF frequency and those below both produce a positive DC voltage which would cancel the result. By grounding one of them, the remaining signal goes up or down, one polarity at a time.

I experimented with the detector outside the receiver by introducing a 6.000 MHz signal from a frequency generator. Once I resonated the maximum 6.000 MHz signal on the secondary winding, I varied the frequency up and down about 100 KHz. I was disappointed that the DC output was only a millivolt or two, positive or negative, but at least it was working. The positive DC voltages were always greater than the negative. I doubted that real signals would produce enough DC changes to make usable audio. You may remember that the receiver in Chapter 7C has a 6.000 MHz crystal which limits the bandwidth to about 3 KHz. In order to detect wideband FM, I believed I needed a wideband signal so I put a shorting switch across the crystal for use in FM mode. I soldered the FM detector into the receiver and set up a test signal using the 2 meter FM signal generator in the transmitter (Chapter 16A). Much to my amazement the audio was beautiful. I found myself enjoying the rebroadcast commercial radio program and forgetting about tweaking the receiver and converter. Gosh! If I had known it would work that well, I would have started with a clean PC board instead of using scrap!



When I first got the 2 meter receiver working I checked into my ham radio club's 2 meter "round table" net using my Icom hand-held. I transmitted using the ground plane vertical antenna up on the roof. I turned on the homebrew receiver with its little foot-high wire antenna. I attached a frequency counter to the VFO output in back so I knew precisely where to tune it. The repeater signal was overwhelmingly strong. In wideband mode (6 MHz crystal shorted) I had difficulty adjusting the HF receiver so that the 146 MHz audio was not badly distorted or breaking into oscillation. Once it was adjusted, I turned off the audio on the handheld and just listened on the homebrew. Fun! The *next week* I wanted to substitute the Icom completely. The dream continued.

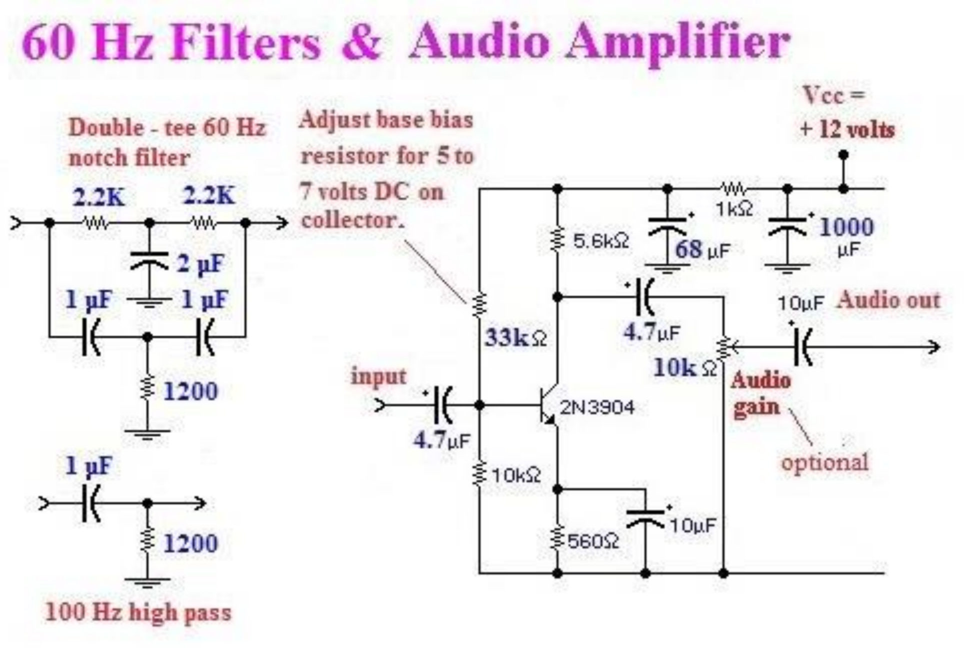
After lots of experimenting, I decided that 2 meter **voice** reception works best with the Chapter 7C receiver set to ***the FM Foster Seeley detector, the HF RF pre-amplifier ON, the 3 KHz single crystal bandwidth, medium level IF gain and relatively high audio gain***. When I use the wideband, no-crystal setting, music can be high fidelity but hams can't legally transmit music. In the wideband mode all the settings are extremely critical and the IF and audio gains must be tuned as carefully as the frequency. In wideband mode a two stage IF amplifier often has too much gain and reception is easily overwhelmed by noise.

AM detection with the Foster-Seeley

A virtue of FM modulation is supposed to be that it is immune to AM noise. I discovered that ***the Foster-Seeley detector receives AM signals extremely well*** and is much more sensitive, i.e., louder, for AM reception than any AM detector system I have tried. Just turn off the BFO, switch to the FM detector and AM voice and music are loud and clear. I assume a "limiter circuit" like the one used in the discriminator FM detector in Chapter 13B could correct the sensitivity to AM noise and modulation.

60 Hz hum

An annoying challenge of FM radio is that ***all*** changes in the signal frequency will be detected. This includes the 60 Hz ripple riding on the VFO power supplies of both the receiver and the transmitter it receives. I was able to improve the hum heard on FM phone reception by changing the 100 µFd cap on the Chapter 7C receiver VFO supply to 6,000 µFd. I also reduced hum by passing the detector audio output through a 1 µFd /1200 ohm R-C filter. This filter can be described as a 100 Hz high pass filter.



Unfortunately, the loss of signal strength due to the 1200 ohm resistor across the output needed an audio amplifier follower to compensate. When using the Chapter 7C receiver, I almost never hear any hum when using the product detector or the simple AM detector. Hum is primarily an FM difficulty. Consequently, I just put the filter and amplifier on the output of the FM detector. It doesn't filter all the inputs to the speaker/ headphone amplifier.

Voice communication is adequate even with the bottom 300 Hz filtered out of the audio spectrum. However, if you want to hear booming bass notes in music, you need a "notch filter" that will preserve the frequencies below 60 Hz. The double-Tee filter shown on the upper left can do this.

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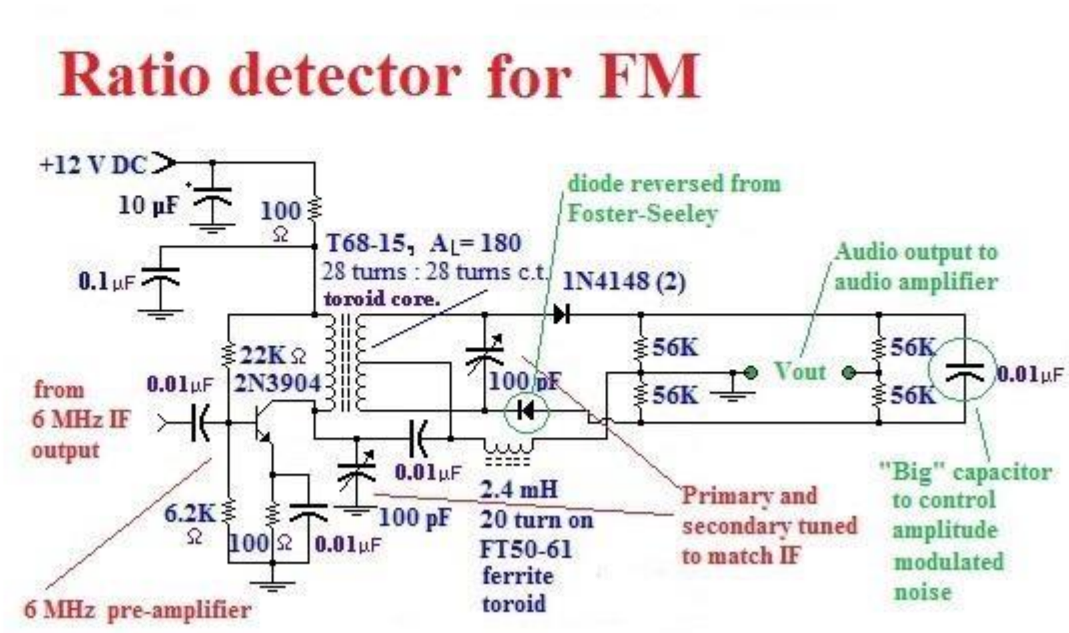
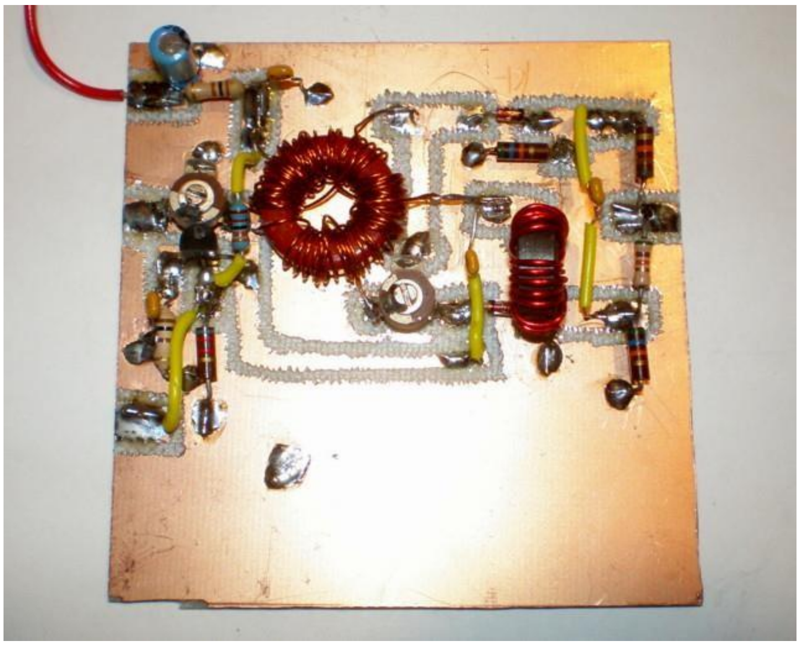
A ratio detector

Because of my difficulties tuning in real ham 2 meter signals, I thought the RF preamp portion of the FM detector might be overloading the Foster-Seeley detector. Also, I had read that a ratio detector has only half of the sensitivity of a Foster-Seeley, but it has other advantages. It is supposed to have a broader bandwidth and better AM noise suppression. I did notice that tuning the 100 pF capacitors was not very critical. I was originally discouraged by most examples of ratio detector designs because they showed all three coils wound on the same core. Keeping track of two 28 turn windings and a center tap was difficult enough.

Then I stumbled across a ratio detector design that was nearly the same as the Foster-Seeley but had a separate RF choke which is easier to build. Also, most ratio detectors are drawn with the output portion of the circuit separate and below the rest of the circuit which makes it *look* more complicated. The circuit below is practically the same as the Foster-Seeley. When you examine it closely, the only big difference is that the two diodes are oriented opposite to each other. Instead of both halves of the winding reinforcing a positive signal, like a full-wave rectifier, the ratio detector seems to deliberately cancel some of the signal. This reversed diode feature seems to be common to all ratio detector diagrams that I found.



The capacitor on the far right is another departure from Foster-Seeley. I read that it is supposed to be "large" to limit the amplitude of the output and suppress amplitude modulated noise. I used a small capacitor for the limiter because I doubted I would prefer less signal output. I decided I could always add capacitance if there were too much atmospheric noise. I installed the detector in the receiver with no 6 MHz preamp into the receiver from Chapter 7C. It was barely working - hardly any output. I added a 6 MHz preamplifier which produced adequate output, but was far less sensitive than the Foster-Seeley. Along the way I learned that some of my oscillations and instability difficulties were caused by an intermittent IF gain potentiometer in the HF receiver. I replaced the pot and tuning became easier. At this stage of development the biggest difficulty with the tuning was that the receiver converter was unshielded and sitting open on top of the enclosure. Where I was happened to sit and how I moved around became a big factor determining whether the little wire antenna had access to the signal. Obviously I needed to put the converter in a metal box and connect it to an outside antenna. That change gave me optimum reception.



The ratio detector board

After checking out the ratio detector, I decided that I preferred the Foster-Seeley detector.

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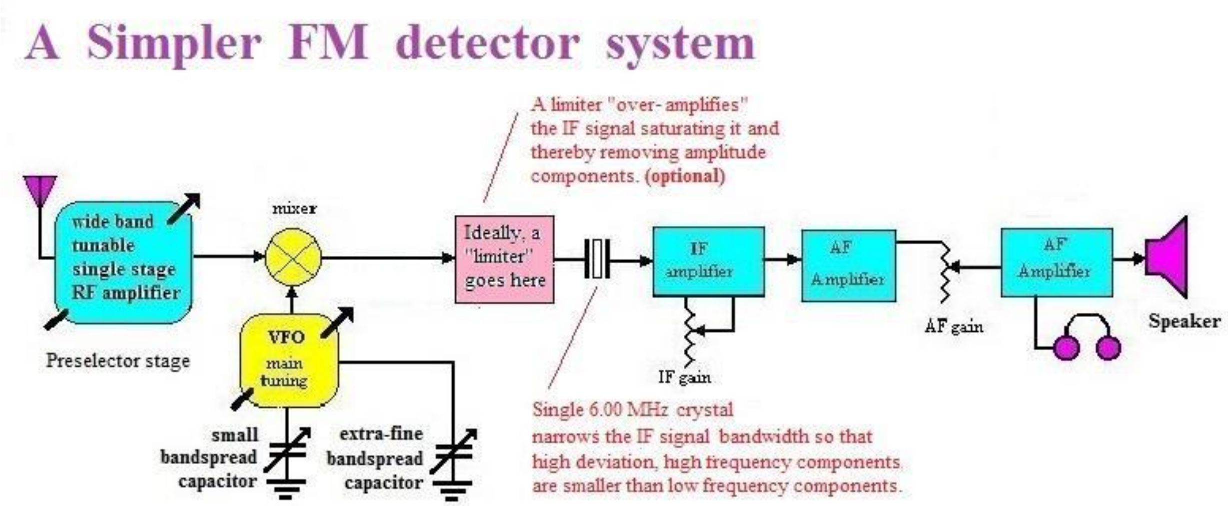
The simplest wideband FM detector

There is nothing like homebrewing to discover new concepts. Well, they're ideas that are at least new to me. Recall that I noticed that AM detectors don't need rectifiers and RF chokes in order to extract the audio amplitude changes from an RF signal. All that is necessary is to feed the IF amplifier output directly into the audio amplifier. The IF signal RF is modulated with the amplitude changes that occur at audio frequencies. The audio amplifier can only amplify the audio frequencies so these are passed along to the loudspeaker.

You will notice that the Foster-Seeley detector has an additional RF amplifier so the IF signal presented to the audio amplifier is much larger than the signal from the IF amplifiers alone. This also produces a much larger audio signal riding on the RF so the signal delivered to audio amplifier is larger and louder.

Wideband FM detection with an AM detector

At the beginning of Chapter 16A I described detecting the original 18 MHz ***narrow-band*** FM signal using the ordinary product detector in the Chapter 13B receiver. The FM deviation adjustment in the transmitter determined whether the audio tone was accurate. When adjusted incorrectly in the transmitter, the voice transmission was either too treble or too bass. When I switched to the AM detector, I could only hear a silent carrier blocking out the atmospheric static. There was no modulation audible. Because the amplitude of an FM signal doesn't vary, there were no amplitude variations for the audio amplifier to amplify. This illustrates why narrowband FM is immune to ordinary AM static noise.



A different kind of crystal set!

In contrast, A ***wide-band*** FM signal can be detected by an audio amplifier provided that the frequency deviation is larger than the nominal bandwidth of the IF signal. The Chapter 7C receiver has a single 6.00 MHz crystal which confines strongest signal components to about 3 KHz. A wide-band IF signal is about 15 KHz wide, so the signal amplitude falls off rapidly with higher deviations away from the 6.00 MHz center frequency. After I rebuilt the audio amplifier, it was capable of much more gain than before. Before this change, I couldn't hear any anything in the AM mode when listening to wide-band FM signals. That situation changed when the audio amplifier became more sensitive to very low levels of input signal.

When the wide-band signal passes through the quartz crystal, the higher deviation components miss the center of the 6 MHz crystal passband and are attenuated. In effect, ***the sharply tuned crystal converts the signal into varying amplitudes proportional to frequency***

- FM to AM conversion. A different kind of crystal set! If you look back at the quartz crystal discriminator I used in Chapter 13B, this detector was operating on a principle quite similar to the way this "accidental" FM detector works.

Notice that this FM detector emphasizes low frequency audio components and diminishes high frequency audio components. The result is an audio signal with big bass response, but much lower high frequency response. When I switch back and forth between AM and Foster- Seeley FM, this AM and FM detector is more pleasant to my ears. You might disagree.

The disadvantages of this detector

A drawback of this detector is that it has no atmospheric noise suppression at all. Just like the crystal sets in Chapter 4, this "crystal set" will pass along any atmospheric or thermal AM noise not covered up by a stronger FM signal. The cure for this problem is (probably) a so- called "limiter circuit." The discriminator circuit in Chapter 13B used a two stage IF amplifier to raise all the AM noise components to saturation level. In theory, the signals presented to the discriminator will just be IF frequency square waves with no amplitude variation. As the IF frequency square waves change frequency, the differing response of the narrow-band crystal will produce amplitude variations. If I manage to get a 6.00 MHz version of the limiter working, I'll add the diagram.

Down-modulation - Loud speech causes inverse amplitude peaks

One aspect of this detector still puzzles me: It ***down-modulates*** voice peaks. The mystery is, why does down-modulation sound so normal? When we speak into the microphone with AM modulation using plate or collector modulation (Chapter 17A) or when using SSB modulation (Chapter 15), the signal power ***rises*** with each syllable when we speak. With this FM detector, speech peaks ***decrease*** the average power to the loudspeaker. Yes, the low frequency components are louder than the high frequency ones, but it sounds OK to me!

Back in 1958 I was using a Heathkit DX-20 CW vacuum tube transmitter. I built a homebrew Heizing cathode AM modulator. This modulator consisted of a large, multi-Henry inductor placed between the cathode and ground. The microphone audio amplifier output was coupled to the cathode. The big inductor prevented the audio from being shorted to ground but allowed the DC cathode current to flow to ground as before. Speech input made the voltage across the final amplifier tube drop instead of rise. When I transmitted voice, the S-meter needle of a receiving station went ***down*** on voice peaks instead of ***up***. It sounded OK, but occasionally the guys I was talking to commented on it.

So tell me again, why do ancient crystal sets need a diode?

In Chapter 7C I realized that "detecting AM" was simply a matter of feeding AM modulated RF into an audio amplifier. Since the only low-frequency components in the RF signal are audio, those are the only frequencies passed onto the loudspeaker. In old-time crystal sets - with galena detectors, etc. - the headphones were always high-inductance electromagnets that caused a steel diaphragm to vibrate. RF currents will not flow through a high inductance. The old headphones were acting like radio frequency chokes that prevent current flow. By converting the RF to changing DC currents, the headphones can respond. Alternatively, the RF can be low-pass filtered so that all that remains are the varying low frequency components. The headphones can respond to low frequency AC. We can think of the diode detector as an extremely simple "low pass filter" rather than some kind of mystical "detector."

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Project status

As of this writing the 2 meter receiver is reliable and working well. Its only fault that still annoys me is that the Chapter 7C receiver VFO drifts. Every minute or two the 2 meter reception frequency drifts a KHz away from the optimum setting. Fortunately wide band FM speech is still intelligible with a 3 or 4 KHz error. A touch on the fine bandspread knob corrects the setting. For instance, when listening to the local repeater, after a minute the 146.700 MHz setting usually drifts down to 146.699 MHz. That's one part out of 147,000. That’s a small error if you look at it that way. It's barely possible that, if I spend more hours adjusting the present VFO circuit, I might be able to improve its stability. That's unlikely with an analog HF VFO oscillator that spans most of the HF spectrum.

As you've probably noticed, older FM broadcast table radios have an FM VFO signal locking circuit. I wasn't the only one with this difficulty. Non-digital, FM broadcast radios use a varactor variable capacitor with a feedback loop to correct minor drifting automatically. My attempts to build locking circuits didn't work and the circuits on the internet use magical ICs or complex circuits I didn't understand. The basic challenge is that the circuit needs to correct for ***both*** rising and falling frequency.

As you can see, VHF communication is not just power amplifiers. It seems endlessly complicated. No solution arrives without introducing a new problem. From my early experiments in UHF, higher frequencies seem to be even worse. Remain calm and enjoy the mysteries! As a boss and mentor of mine once told me, "A great engineer is a persistent engineer."